

House Rental System

HOUSE RENTAL SYSTEM
JIM, SHANJIDA SULTANA

ABC | 01XXXXXX

Client

XYZ Limited

Service Provider

ABC IT Solutions

Project Overview

An Online Room Rental System will provide the Information about Rooms/Flats/Houses which is available for Rent. The aim & the objectives of our project are to create an app for the people and they can rent houses/flats using this app. Room Rental System application will make easy to find the location of Rooms/Flats/Houses, select no. of rooms and other facts by the renter. It will make easy to upload the location, contact No., expected rent, No. of rooms, Facilities and other information by landlord/room-master. This application will be able to show the rooms/flats/houses with in a particular area selected by user. This application will be helpful for the students also. In this pandemic situation we all are working from home. Schools, colleges and universities students are continuing their studies through online. But a big problem will be raised when this pandemic will be gone and they have to comeback outside of their own cities to find flats. Although finding a flat is already hard so it will be harder when this pandemic will be gone and they will need housing for their needs. So, we came up with a plan that can help them to easily find their flat/house. From our app they could easily compare living space & also the facilities within their budget range without visiting the place itself so, that can be time efficient for students and they can take their decision easily.

Justification

House rental system has become important factor in modern society hence the need to have a rental house management application. It is difficult to find house for the renter in a specific area. It is also difficult to find the renter on time, for the landlord and property managers. Recently This Kinds of system working in New York, Boston, or United States. Study of currently available open source applications developed in foreign countries. Analysis the applications and try to modify them according to Bangladeshi environment. Try to develop an efficient application that will be helpful for not only renters but also landlords. Application development will be done according to requirement of both renters and landlords.

Goals

- ✚ To produce a web-based system that allow customer to register and reserve houses online and for the company to effectively manage their House rental business.
- ✚ To Transform the manual process of renting a house to an online and computerized system.
- ✚ Application will be a service which will connect Renters to Landlords and vice- versa.
- ✚ it will save the physical hard work and invaluable time to find room/flat.
- ✚ It will be an Open Source Application which can be freely Install in Android Smart Phones.

Constraints

- ✚ This online house rental solution is fully functional and flexible.
- ✚ Quick deal.
- ✚ The software acts as an office that is open 24/7.
- ✚ Direct communication between renter and landlord.

Assumptions

House Rental is a house/Apartment/home that can be used temporarily for a fee during a specified period. The individual who needs a house must contact a House Rental Owners Through House rental Online System by first checking the available Houses, booking it then The system will contact the house owners to check the House and contract out for a room/home/apartment. This system increases customer retention and simplify House and staff management. The System can be used by Real Estate Companies to increase the house rental market around the world. The system will also help people to rent apartments/houses easily instead of walking and manually renting houses. It is also applicable for landlords who wants to expand their house rental business.

Project Duration

12 weeks

Project Start Date

Date here will be the date that the advance payment is cleared.

Modules and Milestones/Phases with cost

Phases	Modules	Description	Duration	Cost (BDT)
Phase 1	Project Management	In this first phase, project management module will be constructed so that the client can start managing their project.	4 weeks	50k
Phase 2	Registration option	In this phase Registration option will be added, customer can register and choose house for rent in this module.	1 week	25k
Phase 3	Login option	In this phase login option will be added, customer can login from this module.	1 week	50k
Phase 4	Search option	In this phase search option will be added, customer search house for rent in this module.	1 week	25k
Phase 5	Booking system	Booking system option will be added, customer book house for rent in this module.	1 week	50k
Phase 6	Payment system	Payment system will be added in this module.	1week	25k
Phase 7	Project development	Full project development will start and all reports will be implemented.	3 weeks	75k
Phase 8	Delivery	User acceptance testing and delivery.	12 weeks	300k

Payment Plan

- ✚ An advance payment of BDT 40,000 will be required to start the project.
- ✚ Rest of the money must be paid based on phases.

Other Terms and Conditions

- ✚ The Advance payment becomes non-refundable after two weeks of the project initiation.
- ✚ After the delivery the client have report all the problems in between 1 months and it will be free of cost. if they inform us after that time period they have to pay for that.

Agreement

On behalf of the product/service provider <i>JIM, SHANJIDA SULTANA</i> CEO ABC IT Solutions	On behalf of the client seeking product/service <i>Robert Bob</i> XYZ Limited
---	--

Name: JIM, SHANJIDA SULTANA

ID: 19-40621-1

Project Title: House Rental System

EXERCISE - 2: Requirement Analysis part 1

Objective: Develop a Specification Document of the Proposed Solution

Procedure:

1. Prepare a standard Software Requirements Specification (SRS) document that specifies the System's Functional Requirements, Non-Functional Requirements, and Project Development Constraints.
2. Your Specification document must address the following questions:

Evaluation: Your specification document must address the following questions:

1. Do the functional requirements demonstrate the functionality of the software? Do the functional requirements have been leveled accordingly? Do the functional requirements have been prioritized? (minimum 5)
2. Do the non-functional requirements demonstrate the qualities of the software? Do the non-functional or quality requirements are quantifiable measure? Do the non-functional requirements have been prioritized? (minimum 5)
3. Do the project requirements demonstrate the system development constraints, also prioritized, and leveled appropriately?

Functional Requirements

- ✚ Software login- The software shall allow users to login with their given username and password.
- ✚ Customer's registration- The system should allow new users to register online.
- ✚ Online reservation of House- Customers should be able to use the system to make booking and online reservation.
- ✚ Automatic update to database once reservation is made or new customer registered- Whenever there's new reservation or new registration, the system should be able update the database without any additional efforts from the admin.
- ✚ Feedbacks to customers- It should provide means for customers to leave feedback.

Priority Level: High

Precondition: user have valid user id and password

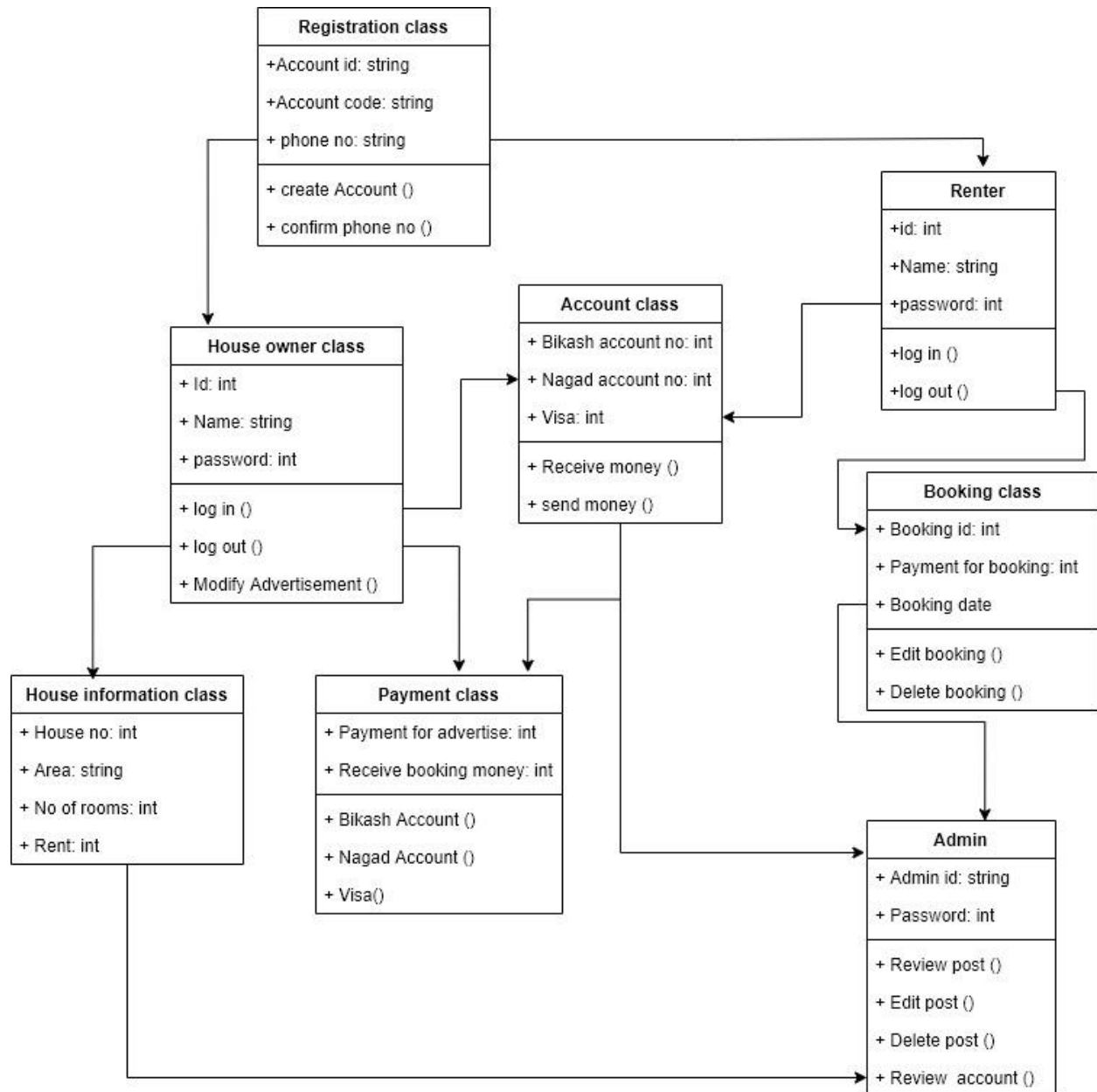
Non-Functional Requirements

- + **Usability:** A trained user shall be able to submit a complete request for house/flat/room within 1 or 2 minutes.
- + **Security:** The subsystem should provide a high level of security and integrity of the data held by the system, only authorized personnel of the company can gain access to the company's secured page on the system; and only users with valid password and username can login to view user's page.
- + **Performance and Response time:** The system should have high performance rate when executing user's input and should be able to provide feedback or response within a short time span usually 5 minutes for highly complicated task and 1 to 2 minutes for less complicated task.
- + **Error handling:** Error should be considerably minimized and an appropriate error message that guides the user to recover from an error should be provided. Validation of user's input is highly essential. Also the standard time taken to recover from an error should be 15 to 20 seconds.
- + **Availability:** This system should always be available for access at 24 hours, 7 days a week.

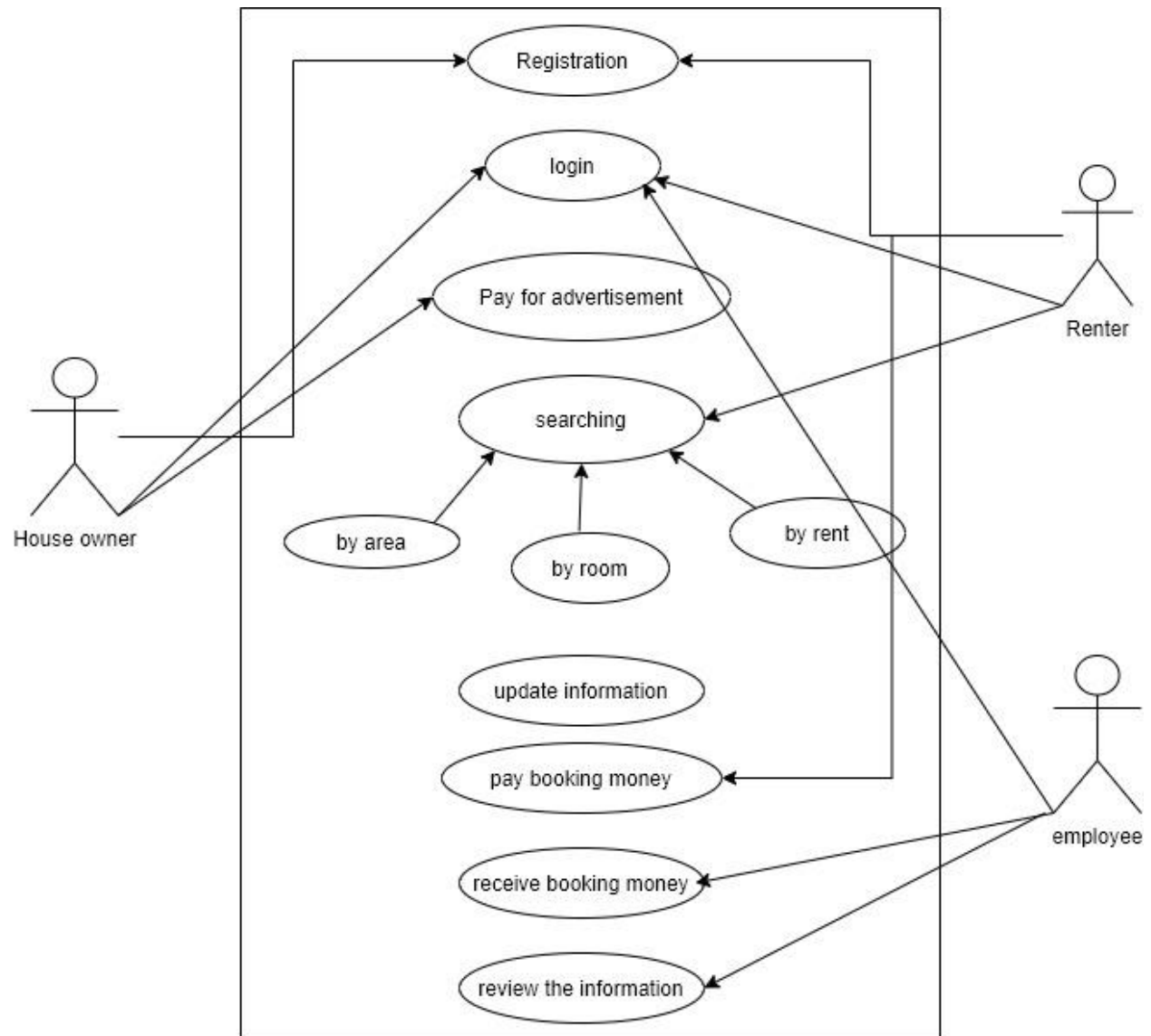
Project Requirements

Tools: The system developer needs DIA tool, GitHub, Trello, SQL, Google map services.

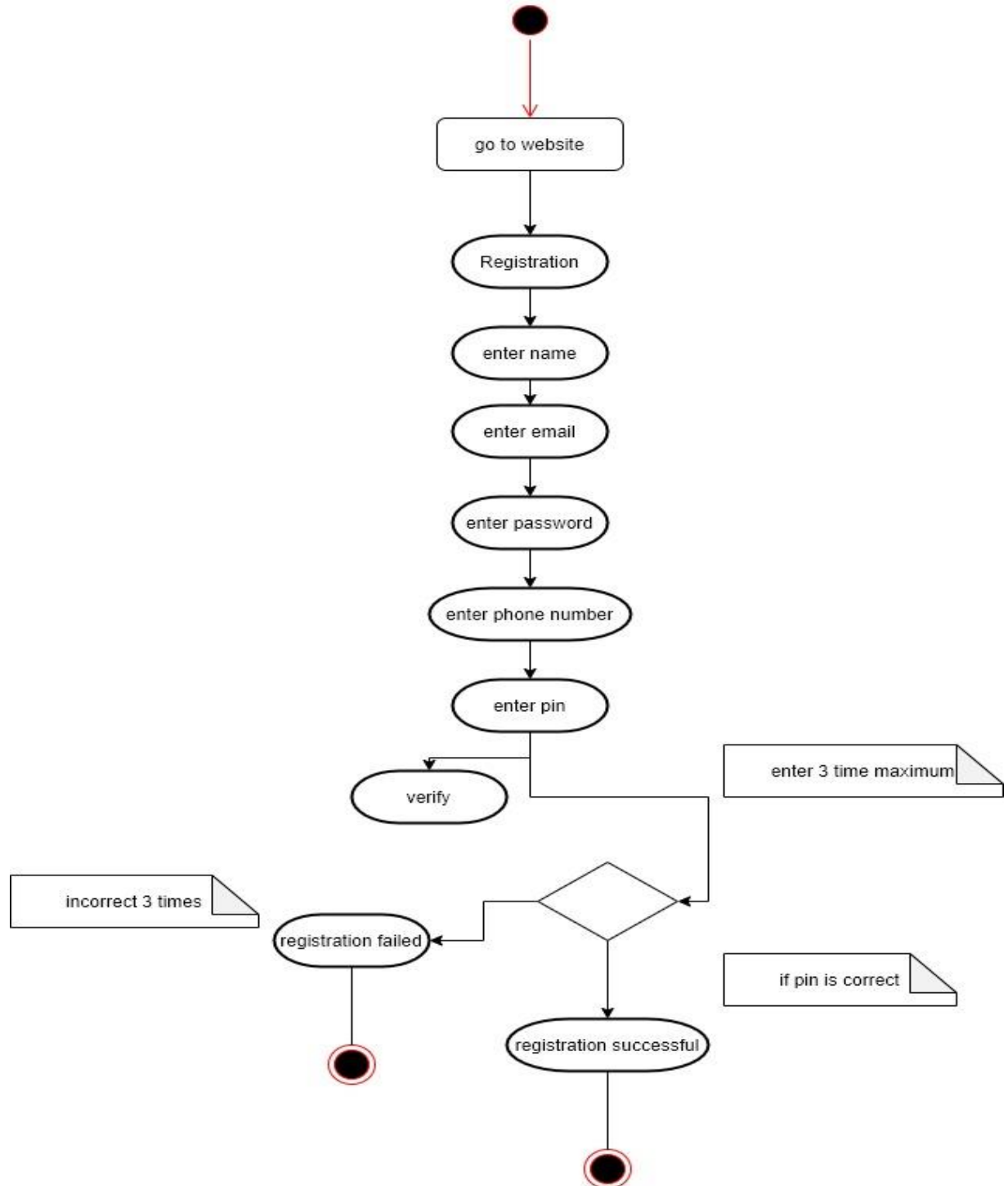
Class Diagram



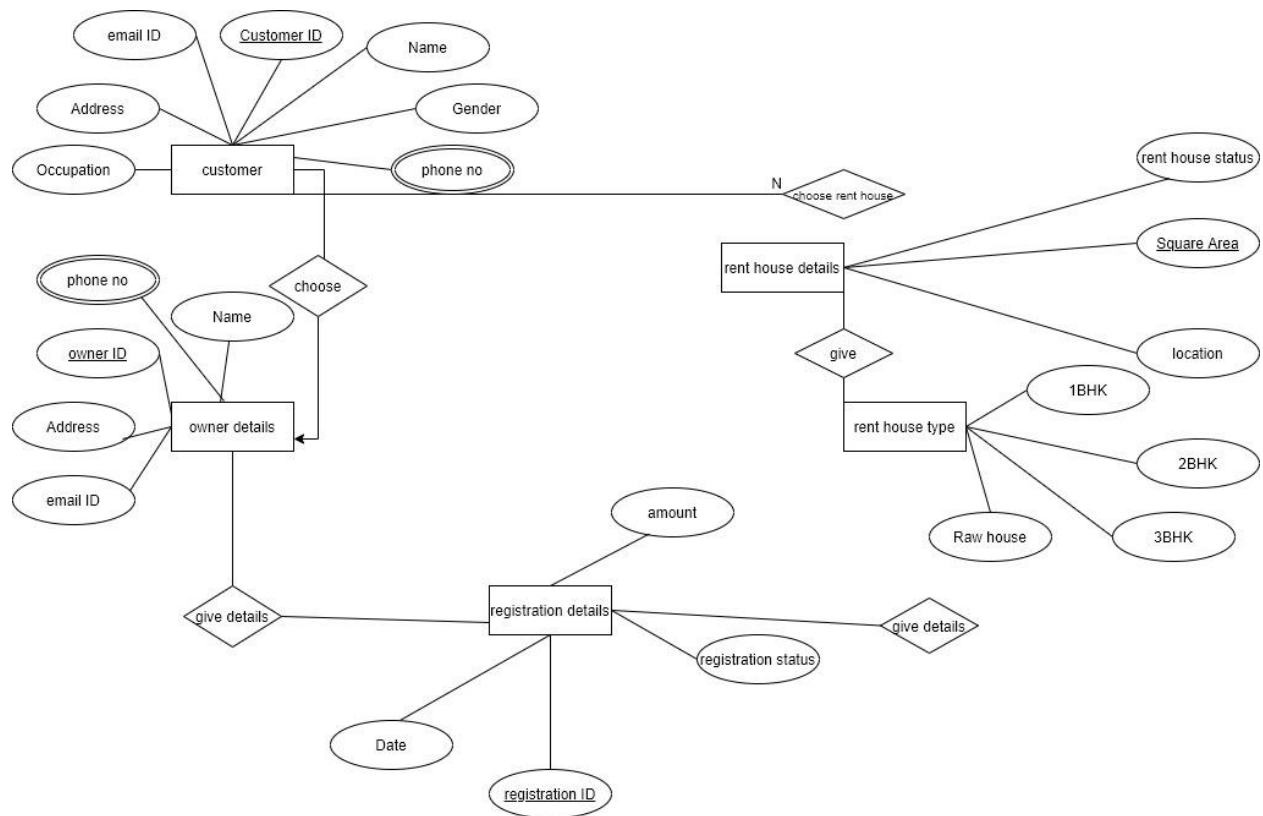
Use Case Diagram



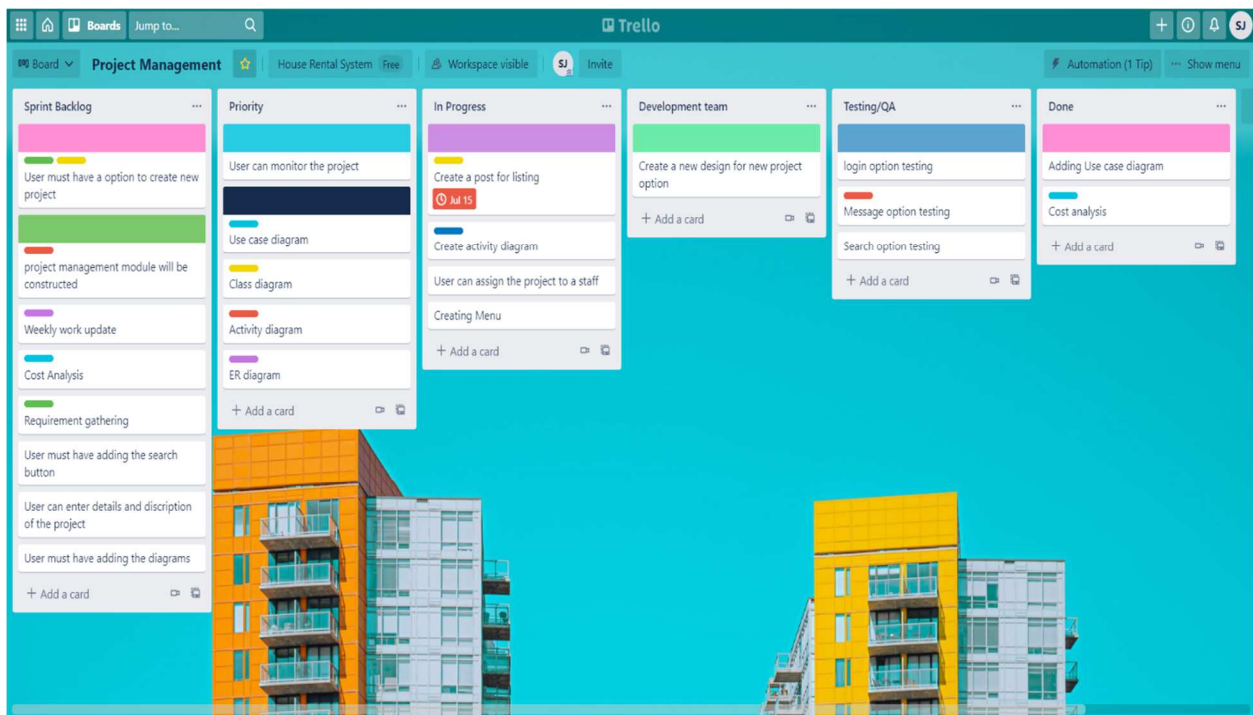
Activity Diagram



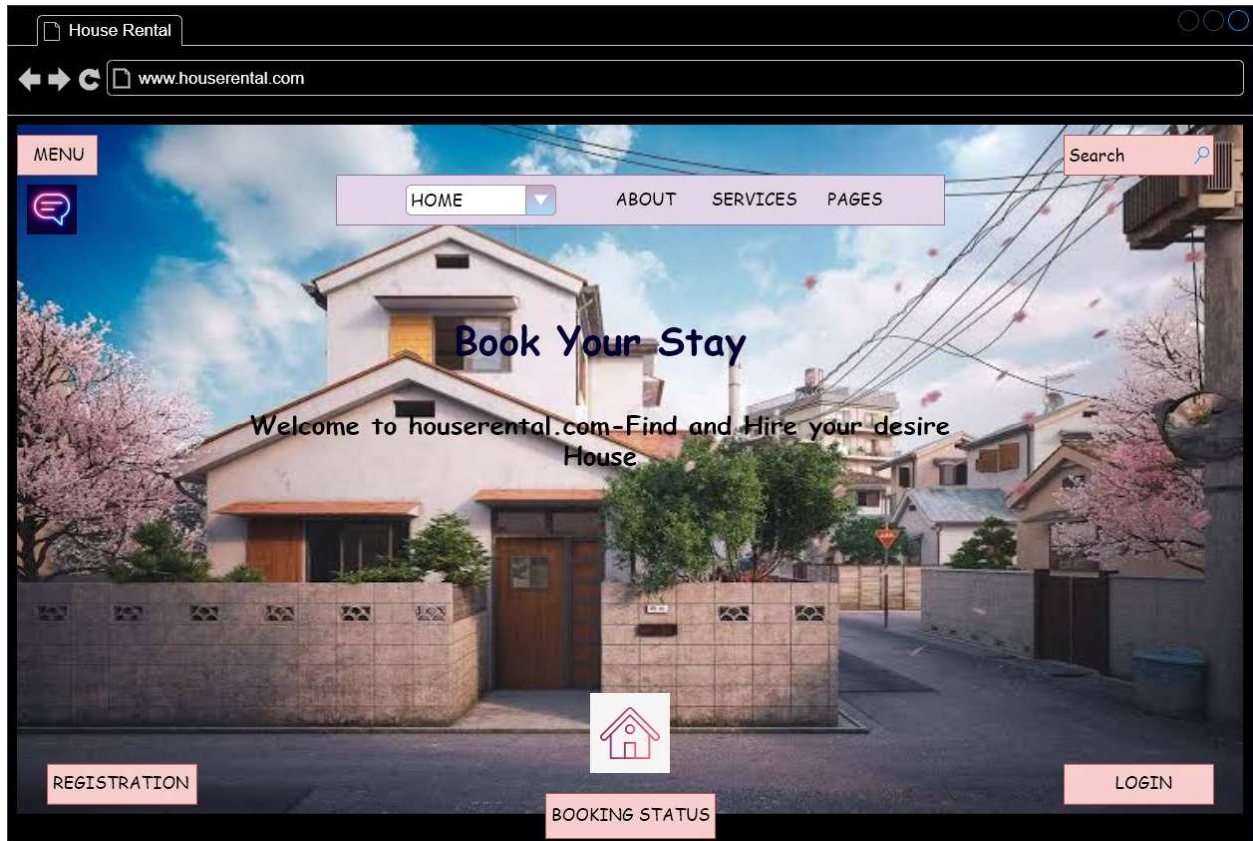
ER Diagram



Trello



UI Mockup using (draw.io)



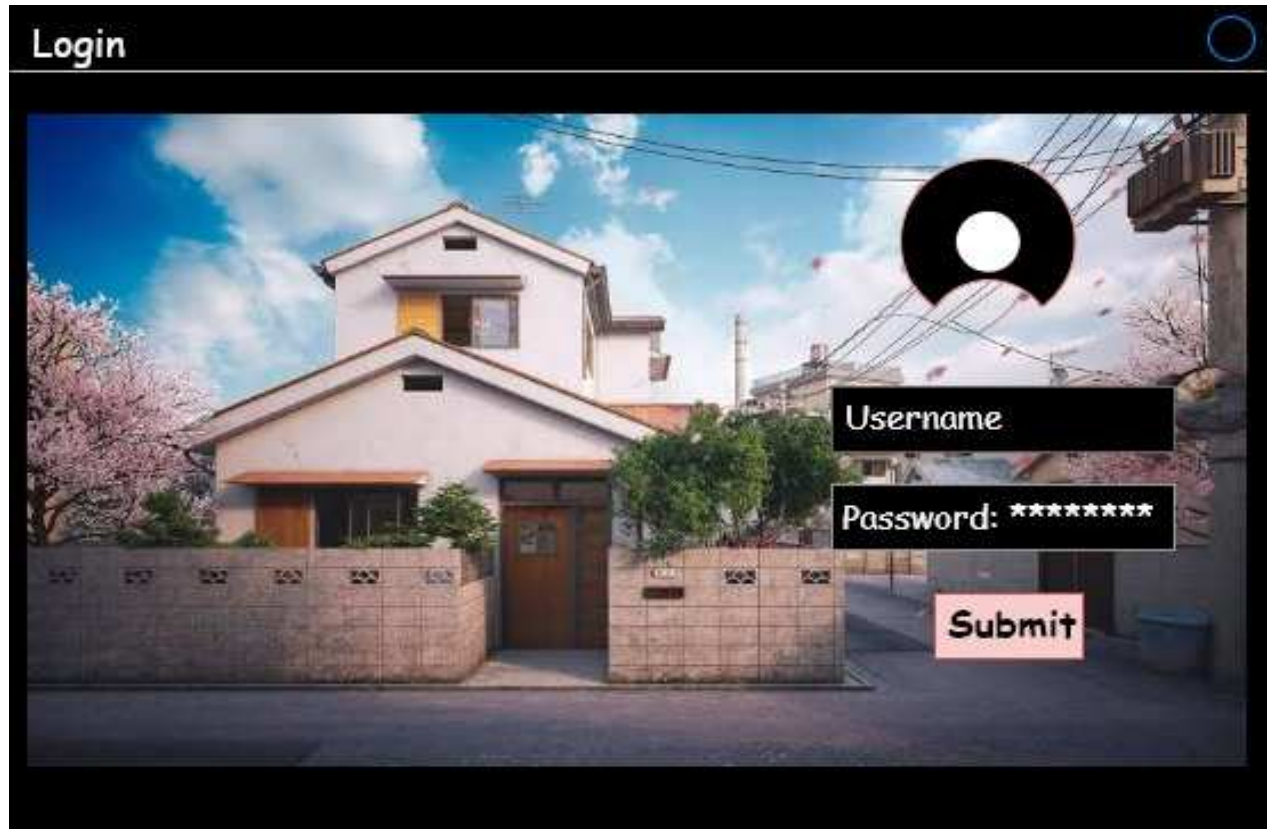
Main Webpage. The house owner open the website and the portal shows the login and registration form for access. If user is not a registered member they can choose registration option and go to the registration menu. Already registered members can choose the login option for login and can check their booking details by choosing the booking status option.

Registration



Register

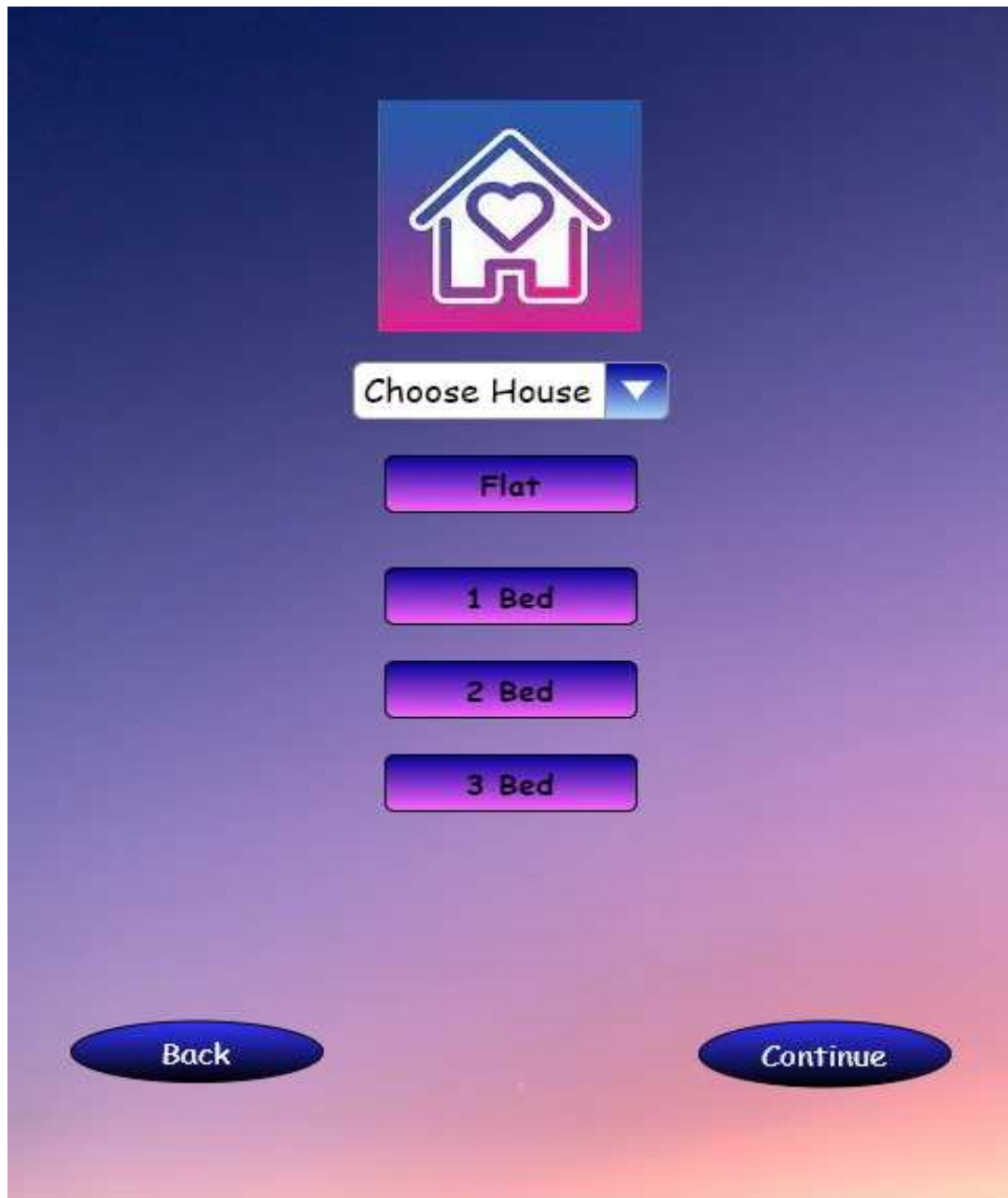
Registration page. New people can register here for searching their desire house.



This is the login page for registered members. House owner's and renter's can login form this page.



This is the payment system page. Here registered customer can select their payment method and pay the rent of house.



This is the booking system page. Here users can select their desire house.

TEST CASE

Project Name: House Rental System		Test Designed by: Jim, Shanjida Sultana		
Test Case ID: HRS_1		Test Designed date: 11/07/2021		
Test Priority (Low, Medium, High): Medium		Test Executed by: Robert Bob		
Module Name: Registration option		Test Execution date: 13/07/2021		
Test Title: New users can register here				
Description: Test website registration page				
Precondition (If any): Users must have submit their verification pin				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Choose registration option 3. Enter name 4. Enter email 5. Enter password 6. Enter phone number 7. Enter verification pin 8. Click register	Name: Elina Email: elina32@gmail.com Password: 321 Phone: 01***** Pin: 0243	User should register into the application	As expected,	Pass
Post Condition: User is validated with database and successfully register to account. The account session details are registered in the database.				

Module Name: Login option		Test Execution date: 13/07/2021		
Test Title: Users can login here				
Description: Test website login page				
Precondition (If any): User must have valid username and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Go to the website 2. Choose login 3.Enter username 4.Enter password 5.Click submit	Username: Elina375 Password: 456*****	User should login into the application	As expected	Pass
Post Condition: User is validated with database and successfully login to account. The account session details are logged in the database.				

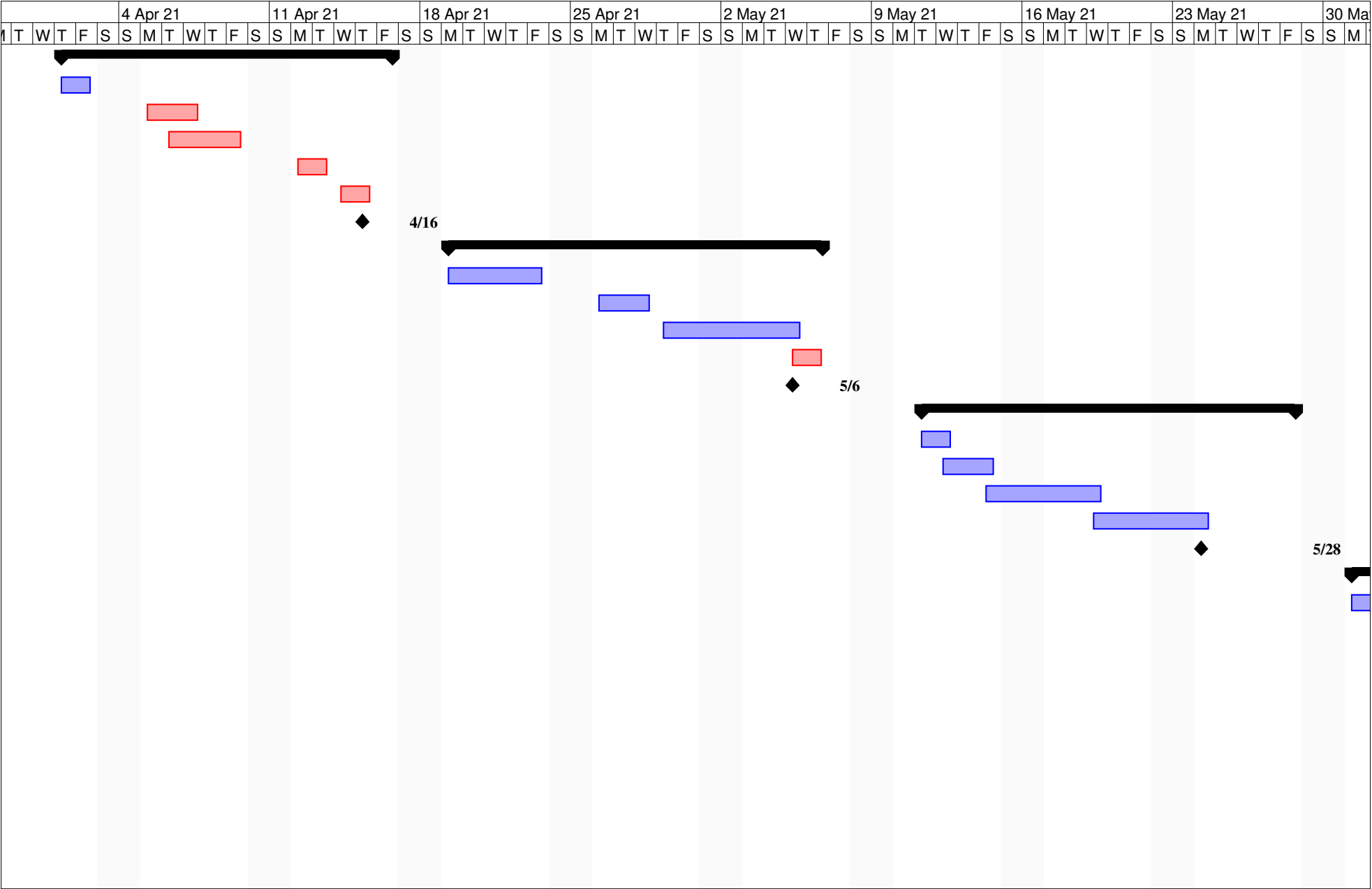
Module Name: Search option		Test Execution date: 13/07/2021		
Test Title: Users can search here				
Description: Test website search option				
Precondition (If any): Search by area				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Go to the website 2. Choose search option 3.Enter your area 4.Click search	Submit area	User should search their desire house into the application	As expected	Pass
Post Condition: User is validated with database and successfully search their desire house.				

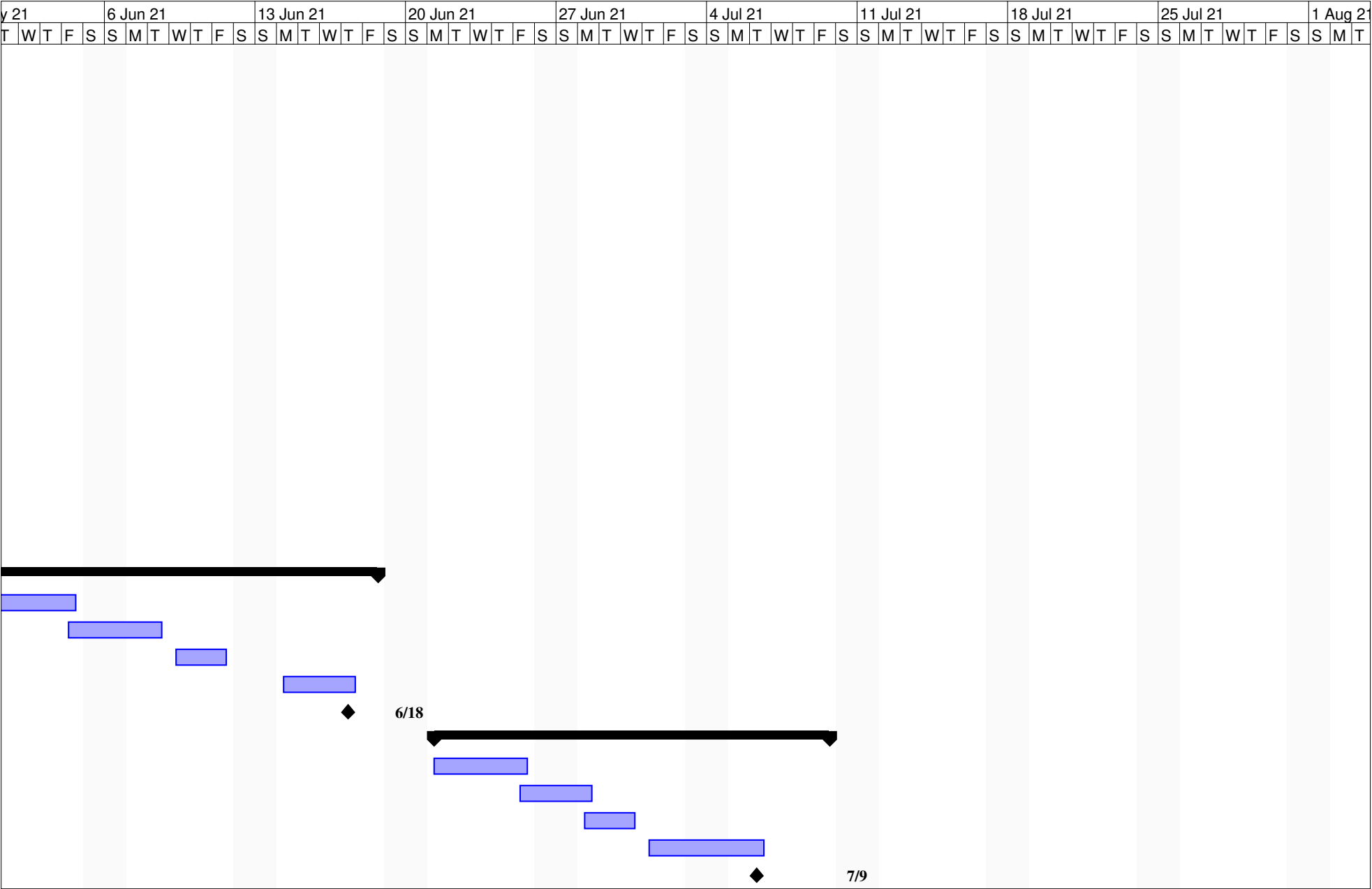
Module Name: Booking system		Test Execution date: 13/07/2021		
Test Title: Users can book house and see the details about booking here				
Description: Test website booking system				
Precondition (If any): Users must be a registered user				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Go to the website 2. Choose booking status 3.See the booking details	Booking house	Users can see the booking status into the application	As expected	Pass
Post Condition: User is validated with database and successfully watch booking status.				

Module Name: Payment system		Test Execution date: 13/07/2021		
Test Title: Users can pay the house rent here				
Description: Test website payment system				
Precondition (If any): Users must be a registered user				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Go to the website 2. Choose login 3.Choose payment option 4. Select payment method 5.Enter amount 6.Complete payment	Payment method: (Bikash, Rocket, Nagad) Enter amount: ***** Enter pin: 67**	Users can pay the rent successfully into the application	As expected	Pass
Post Condition: User is validated with database and successfully pay the rent.				

GIT HUB: [https://github.com/Shanjida03/SE SUMMER 2021](https://github.com/Shanjida03/SE_SUMMER_2021)

		Name	Duration	Start	Finish	Predecessors	Resource Names	N
1		Project management	12 days?	4/1/21 8:00 AM	4/16/21 5:00 PM			
2		Project research	2 days?	4/1/21 8:00 AM	4/2/21 5:00 PM			
3		Understand your stakeh...	3 days?	4/3/21 8:00 AM	4/7/21 5:00 PM			
4		Understand your own role	4 days?	4/6/21 8:00 AM	4/9/21 5:00 PM			
5		Set a project vision	2 days?	4/10/21 8:00 AM	4/13/21 5:00 PM			
6		Set a project goal	2 days?	4/14/21 8:00 AM	4/15/21 5:00 PM			
7		Milestone-Project planning	2 days?	4/15/21 8:00 AM	4/16/21 5:00 PM			
8		Registration	14 days?	4/19/21 8:00 AM	5/6/21 5:00 PM			
9		Create plan	5 days?	4/19/21 8:00 AM	4/23/21 5:00 PM			
10		Create UI layout	3 days?	4/24/21 8:00 AM	4/28/21 5:00 PM			
11		Make all your fields requi...	5 days?	4/29/21 8:00 AM	5/5/21 5:00 PM			
12		Make password requirem...	2 days?	5/5/21 8:00 AM	5/6/21 5:00 PM			
13		Milestone-Registration p...	2 days?	5/5/21 8:00 AM	5/6/21 5:00 PM			
14		Payment System	14 days?	5/11/21 8:00 AM	5/28/21 5:00 PM			
15		Disscuss with clients	2 days?	5/11/21 8:00 AM	5/12/21 5:00 PM			
16		Design Planning	3 days?	5/12/21 8:00 AM	5/14/21 5:00 PM			
17		Create payment methods	4 days?	5/14/21 8:00 AM	5/19/21 5:00 PM			
18		Customer privacy support	4 days?	5/19/21 8:00 AM	5/24/21 5:00 PM			
19		Milestone- Completed pa...	5 days?	5/24/21 8:00 AM	5/28/21 5:00 PM			
20		Booking System	15 days?	5/31/21 8:00 AM	6/18/21 5:00 PM			
21		Customized booking page	5 days?	5/31/21 8:00 AM	6/4/21 5:00 PM			
22		Search house	3 days?	6/4/21 8:00 AM	6/8/21 5:00 PM			
23		Select location	3 days?	6/9/21 8:00 AM	6/11/21 5:00 PM			
24		Test	4 days?	6/12/21 8:00 AM	6/17/21 5:00 PM			
25		Milestone-Completed bo...	2 days?	6/17/21 8:00 AM	6/18/21 5:00 PM			
26		Project Planning	15 days?	6/21/21 8:00 AM	7/9/21 5:00 PM			
27		Identify and plan the av...	5 days?	6/21/21 8:00 AM	6/25/21 5:00 PM			
28		Set up a communication ...	2 days?	6/25/21 8:00 AM	6/28/21 5:00 PM			
29		Work on the budget	3 days?	6/28/21 8:00 AM	6/30/21 5:00 PM			
30		Project Monitoring and C...	4 days?	7/1/21 8:00 AM	7/6/21 5:00 PM			
31		Milestone-Development ...	4 days?	7/6/21 8:00 AM	7/9/21 5:00 PM			





[illegible]

END